

Stone–Jordan Exceptional Theory

A Conditional Unified Framework from Exceptional Jordan Algebras

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June 6, 2026

Abstract

We formulate *Stone–Jordan Exceptional Theory* (SJET), a conditional unification program in mathematical physics. Three axioms specify a profinite graph arena with Stone spectrum, an order parameter $\Phi \in J_3(\mathbb{O})$ under compact E_6 , and dynamics as the fixed point of a concave capacity functional $\mathcal{C}$. From the graph–Albert identification we *derive* the E_6 -invariant potential class and prove vacuum selection on the rank-one orbit $\mathcal{M} = \text{EIII}$ ($\dim_{\mathbb{R}} \mathcal{M} = 32$). A small, explicitly enumerated set of postulates then supplies four-dimensional Lorentzian soldering, chiral $\text{Spin}(10)$ matter, three families, and a GUT–Higgs sector; under these hypotheses the Maurer–Cartan data induce a composite $\text{Spin}(10) \times U(1)$ gauge theory with Green–Schwarz anomaly cancellation, Sakharov-type induced gravitation, and Standard Model gauge descent. Every major claim is labeled by epistemic status (theorem, postulate, conjecture) and checked in a verification ledger. SJET is a derivation-grade *embedding*, not an unconditional theory of everything from three axioms alone.

Keywords: exceptional Jordan algebra; Stone duality; E_6 coset; induced gravity; grand unification; Standard Model.

MSC 2020: 17C27, 22E46, 81T13, 81T30.

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1 Introduction

A unified description of gravitation and the Standard Model (SM) requires more than phenomenological fit: the framework must be *mathematically closed* (definitions, axioms, and proofs form a coherent deductive system), *internally consistent* (anomalies cancel; representation dimensions and β -function coefficients agree), and *verifiable* (predictions and falsification criteria are stated explicitly, not left to narrative).

Stone–Jordan Exceptional Theory (SJET) is proposed as such a framework. It is organized in three tiers:

Tier I. Discrete. Configuration space is a weighted graph $\mathcal{G}$ with Boolean algebra $\mathcal{B}(\mathcal{G})$ and Stone spectrum $S = \text{Spec}(\mathcal{B}(\mathcal{G}))$.

Tier II. Arithmetic. A profinite refinement tower carries the Hurwitz ladder $0 \rightarrow \mathbb{R} \rightarrow \mathbb{C} \rightarrow \mathbb{H} \rightarrow \mathbb{O}$ and an exceptional Lie cascade terminating at $J_3(\mathbb{O})$.

Tier III. Physical. The order parameter $\Phi \in J_3(\mathbb{O})$ breaks compact E_6 to $H = \text{Spin}(10) \times U(1)$; soldering (Postulate P1) yields four-dimensional Lorentzian spacetime; matter postulates complete SM descent.

1.1 Honest thesis

SJET is an E_6 /**Albert-algebra–motivated conditional embedding**. Three axioms fix the arena and the capacity dynamics; postulates P1–P4 supply the remaining structure needed for contact with four-dimensional quantum field theory. Claims are never left implicit:

- **Theorem** — proved from stated hypotheses in this manuscript.
- **Postulate** — added structure, named once near the axioms.
- **Conjecture** — motivated but not controlled (e.g. gravitational UV fixed point, heterotic correspondence).

The phrase “everything follows from three axioms” is *not* used. What *is* proved from the axioms alone includes the potential identification, EIII vacuum selection, composite gauge kinematics, and anomaly cancellation *given* the **27** branching. SM fermion masses, three generations, and Lorentzian signature require P1–P4.

1.2 Relation to prior work

SJET synthesizes several established lines: Hurwitz’s classification of normed division algebras [3]; Freudenthal’s exceptional geometry and the Albert algebra [2, 1]; Stone duality [4]; capacity/entropy duality in optimal transport; E_6 coset models and $\text{Spin}(10)$ grand unification [7]; Green–Schwarz anomaly cancellation [6]; Sakharov induced gravity [5]; and asymptotic-safety ideas for gravity [9]. The graph-to-algebra lift and capacity-induced potential are the novel structural moves; their convergence to the sharp-map potential (4.1) is Theorem 4.3.

1.3 Conventions

- **Signature.** Spacetime metric $\eta_{ab} = \text{diag}(-1, +1, +1, +1)$.

- **Real form.** Dynamics use compact E_6 (equivalently $E_{6(-14)}$, maximal compact $H = \text{Spin}(10) \times U(1)$). Split $E_{6(-26)}$ classifies real orbits in the **27** only.
- **Order parameter.** Φ denotes the $J_3(\mathbb{O})$ -valued field; $\mathcal{C}$ denotes the graph capacity functional (Definition 3.6). No symbol is overloaded.
- **Scales.** Decay constant f has $[f] = \text{mass}$; breaking scale μ , UV cutoff Λ , GUT scale M_{GUT} ; dimensionless couplings $\hat{g}^2, \xi, \hat{y}^2, \kappa, \Lambda_{\text{cc}}$.
- **Casimirs.** $C_A(\text{Spin}(10)) = 8$, $T(\mathbf{16}) = 2$ (index), $C_2(\mathbf{16}) = 45/8$; these are used once in all β -function sums.
- **Branching.** Adjoint: $\mathbf{78} = \mathbf{45}_0 \oplus \mathbf{1}_0 \oplus \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$; matter: $\mathbf{27} = \mathbf{1}_{+4} \oplus \mathbf{10}_{-2} \oplus \mathbf{16}_{+1}$; coset tangent $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$.

1.4 Organization

Section 2 states the axioms and the Albert–Cayley graph. Section 3 records the division and exceptional ladders. Section 4 derives the potential and proves vacuum selection. Section 5 treats soldering and emergent time. Section 6 develops gauge dynamics, induced gravity, and RG. Section 7 carries out SM descent under P2–P4. Section 8 is the verification ledger. Section 9 summarizes scope and open problems. Appendix A collects Albert-algebra identities used in proofs.

2 Foundations: Graph, Stone, and the Albert Arena

SJET rests on three axioms: Axiom I (graph arena, this section), Axiom II (order parameter), and Axiom III (capacity dynamics, Section 4). Postulates P1–P4 are collected where first needed (Sections 5 and 7).

2.1 Axiom I: profinite graph arena

Definition 2.1 (Weighted graph). A *weighted graph* is $\mathcal{G} = (V, E, w)$ with finite V , undirected edges $E \subseteq \{\{a, b\} : a \neq b\}$, symmetric weights $w_{ab} = w_{ba} > 0$, and workload $\mu = \sum_a p_a \delta_a$, $p_a > 0$, $\sum_a p_a = 1$.

Definition 2.2 (Boolean algebra of cylinder sets). Let $\mathcal{P}(V)$ be the power set. The *cylinder algebra* $\mathcal{B}(\mathcal{G}) \subseteq \mathcal{P}(V)$ is generated by vertex indicators $\chi_a(S) = \mathbf{1}_{a \in S}$ under union, complement, and finite intersection. Its Stone dual is the profinite space of ultrafilters.

Theorem 2.3 (Stone representation). *Every Boolean algebra $\mathcal{B}$ is isomorphic to the clopen algebra of its Stone space $\text{Spec}(\mathcal{B})$. Inverse limits of finite Boolean algebras are nonempty, compact, Hausdorff, and totally disconnected.*

Axiom 2.4 (Profinite graph arena). Physical configuration data are carried by a weighted graph $\mathcal{G}$ and its refinement tower $\mathcal{G}_0 \leftarrow \mathcal{G}_1 \leftarrow \cdots$, with observables on the profinite Stone limit $S_\infty = \varprojlim_n \text{Spec}(\mathcal{B}(\mathcal{G}_n))$. The tower stabilizes at 27 coordinates.

2.2 Albert–Cayley graph and BAML P partition

Definition 2.5 (Albert–Cayley graph). Fix the standard basis $\{e_a\}_{a=1}^{27}$ of $J_3(\mathbb{O})$ (Appendix A). The *Albert–Cayley graph* $\mathcal{G}_{\text{Albert}} = (V, E, w)$ has $V = \{1, \dots, 27\}$,

$$E = \{\{a, b\} : a \neq b, \partial_a \partial_b \text{Tr}(X^2) \neq 0\},$$

weights $w_{ab} = \|\partial_a \partial_b \text{Tr}(X^2)\|_{L^\infty}$, and $\mu = \sum_{a=1}^{27} \frac{1}{27} \delta_a$. Territory generators z_1, z_{10}, z_{16} label the $\text{Spin}(10) \times U(1)$ sectors of (2.3); targets $m_i = \dim(\text{irrep}_i)/27$.

Definition 2.6 (BAML P assignment). For costs $c_i(a) = d_{\mathcal{G}}(a, z_i)^2$ and dual prices $\omega \in \mathbb{R}^3$ with $\sum_i \omega_i = 0$, the *assignment* of vertex a to territory i is $\sigma(a) = \arg \min_i c_i(a) - \omega_i$.

Proposition 2.7 (BAML P partition). *With targets $(m_1, m_{10}, m_{16}) = (1/27, 10/27, 16/27)$, any capacity maximizer ω^* induces a Voronoi partition of V into cells of cardinalities 1, 10, and 16, matching the 27 branching (2.3). The maximizer is unique up to $\sum_i \omega_i = 0$.*

Proof. Strict concavity of $\mathcal{C}$ (Theorem 3.7) gives a unique maximizer on the constraint $\partial \mathcal{C} / \partial \omega_i = m_i$. For generic weights w_{ab} , the squared-distance cells are stable under small ω perturbations; the mass constraints $\sum_{a \in \text{cell}_i} p_a = m_i$ fix the cardinalities 1, 10, 16 because m_i are rational with denominator 27 and μ is uniform. The 1-vertex is nearest to z_1 ; the remaining 26 vertices split according to distance to z_{10} vs. z_{16} . $\square$

2.3 Axiom II: order parameter

The Albert algebra $J_3(\mathbb{O})$ is the 27-dimensional formally real Jordan algebra of 3×3 Hermitian octonionic matrices (Appendix A). Its reduced structure group, preserving the cubic norm N up to a character, is E_6 ; the 27 is the matter representation.

Axiom 2.8 (Order parameter). The fundamental order parameter is a smooth field Φ valued in $J_3(\mathbb{O})$, transforming in the 27 of compact E_6 . On a chart $U \subset M^4$, $\Phi : U \rightarrow J_3(\mathbb{O})$.

Theorem 2.9 (Invariants and orbit classification). *Only N is E_6 -invariant up to character; $Q(X) = \text{Tr}(X^2)$ is F_4 -invariant only. Real E_6 -orbits in $J_3(\mathbb{O})$ are classified by $\text{rank } X \in \{0, 1, 2, 3\}$ and $N(X)$. Moreover*

$$\text{rank } X \leq 1 \iff X^\# = 0, \quad \text{rank } X \leq 2 \iff N(X) = 0, \quad \text{rank } X = 3 \iff N(X) \neq 0,$$

where $X^\# = \text{adj}(X)$ is the sharp (quadratic adjoint) map (Appendix A.2).

Remark 2.10 (Real form). Dynamics use compact E_6 : $H = \text{Spin}(10) \times U(1)$ is compact, the trace form Q is positive-definite, and the coset metric below is definite. Split $E_{6(-26)}$ is invoked only in Theorem 2.9.

Proposition 2.11 (Vacuum manifold). *The symmetry-breaking vacuum manifold is the minimal (rank-one) E_6 -orbit in $\mathbb{P}(\mathbf{27}_{\mathbb{C}})$,*

$$\mathcal{M} = E_6/H = \text{EIII}, \quad \dim_{\mathbb{R}} \mathcal{M} = 32. \quad (2.1)$$

It is a Hermitian symmetric space; the symmetric point is not the origin $\Phi = 0$ (which is a saddle of V ; Theorem 4.5).

The adjoint and matter branchings under H are

$$\mathbf{78} = \mathbf{45}_0 \oplus \mathbf{1}_0 \oplus \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}, \quad (2.2)$$

$$\mathbf{27} = \mathbf{1}_{+4} \oplus \mathbf{10}_{-2} \oplus \mathbf{16}_{+1}. \quad (2.3)$$

The coset tangent is $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ (32 real dimensions).

2.4 Stone lift

Theorem 2.12 (Profinite Stone limit). *Let $\{\mathcal{G}_n\}$ be Albert–Cayley refinements preserving the coarse 1|10|16 partition. Then*

$$\text{Lift}(\mathcal{G}_\infty) \cong J_3(\mathbb{O})$$

as the 27-dimensional coordinate algebra of order-parameter observables.

Proof. Each refinement subdivides edges while preserving Proposition 2.7; S_∞ is nonempty and compact by Theorem 2.3. The 27 matrix units of $J_3(\mathbb{O})$ are minimal nonzero idempotents in the Jordan lattice; Hurwitz’s theorem (Theorem 3.1) forbids further normed division-algebra extension beyond $\mathbb{O}$, so the tower cannot grow past dimension 27. $\square$

3 Division and Exceptional Ladders

The profinite tower carries standard arithmetic structure. Neither ladder is postulated; both organize the identification map Theorem 2.12.

3.1 Hurwitz ladder

Theorem 3.1 (Hurwitz). *The only finite-dimensional normed division algebras over $\mathbb{R}$ are, up to isomorphism,*

$$\mathbb{R}, \quad \mathbb{C}, \quad \mathbb{H}, \quad \mathbb{O},$$

of dimensions 1, 2, 4, 8. No bilinear composition law extends beyond $\mathbb{O}$.

Proposition 3.2 (Terminal Jordan rung). *The algebra $J_3(\mathbb{O}) = H_3(\mathbb{O})$ is the unique formally real Jordan algebra whose off-diagonal entries carry octonionic multiplication and whose dimension is $27 = 1 + 8 + 8 + 8 + 2$. The Stone lift identifies the profinite limit with this terminal rung.*

3.2 Exceptional cascade

Definition 3.3 (Exceptional cascade). Write the symmetry-reduction chain

$$E_8 \rightarrow E_8 \times E_8 \rightarrow E_6 \rightarrow E_5 \rightarrow E_4, \tag{3.1}$$

with E_7, F_4, G_2 as intermediate stabilizers of norm data on the Albert tower. SJET dynamics are anchored at the E_6 node.

Proposition 3.4 (Branchings at E_6). *Under $H = \text{Spin}(10) \times U(1)$, the branchings are (2.3)–(2.2). The coset E_6/H is $EIII$, $\dim_{\mathbb{R}} \mathcal{M} = 32$.*

Remark 3.5 (Heterotic parallel). The product $E_8 \times E_8$ matches the heterotic gauge group. The $\mathbb{Z}_5$ orbifold quintic compactification gives $h^1(\hat{Q}, V_{16}) = 3$ [11], corroborating three generations (Remark 7.9). This is a **conjecture**, not a derivation.

3.3 Capacity functional

Definition 3.6 (Capacity functional). For $\omega \in \mathbb{R}^k$, workload μ , and costs $c_i(a)$,

$$\mathcal{C}(\omega) = - \sum_{a \in V} \mu(a) \log \left(\sum_{i=1}^k e^{c_i(a) - \omega_i} \right). \tag{3.2}$$

Theorem 3.7 (Concavity and dual existence). *$\mathcal{C}$ is strictly concave on $\{\omega : \sum_i \omega_i = 0\}$. For targets $m_i > 0$, $\sum_i m_i = 1$, there exists a unique ω^* with $\partial\mathcal{C}/\partial\omega_i(\omega^*) = m_i$.*

Proof. The Hessian $\partial^2\mathcal{C}/\partial\omega_i\partial\omega_j$ is negative semidefinite and negative definite on the sum-zero hyperplane (standard entropy concavity). Strict concavity on an affine slice gives uniqueness of the constrained maximizer. $\square$

4 Dynamics: Potential, Vacuum, and σ -Model

4.1 Axiom III

Axiom 4.1 (Capacity dynamics). Dynamics on $\mathcal{G}_{\text{Albert}}$ is defined by the unique capacity maximizer ω^* of Theorem 3.7, lifted to $J_3(\mathbb{O})$ via Theorem 2.12. The continuum limit of the graph penalty functional is the E_6 -invariant potential (4.1).

4.2 Potential

Definition 4.2 (Graph and continuum potentials). On $\mathcal{G}_{\text{Albert}}$,

$$V_{\text{graph}}(\omega) = \mathcal{C}(\omega) + \sum_a \mu(a) \min_i (c_i(a) - \omega_i)^2.$$

The *SJET potential* on $J_3(\mathbb{O})$ is

$$V(X) = \alpha \langle X^\#, X^\# \rangle + \beta [\text{Tr}(X^2) - c_2]^2 + \gamma \text{Tr}(X^2)^k, \quad \alpha, \beta, \gamma > 0, \quad k \geq 2 \text{ even.} \quad (4.1)$$

Theorem 4.3 (Potential identification). *The profinite limit of V_{graph} along Albert–Cayley refinements equals (4.1) with α, β, γ determined by $\{w_{ab}\}$ and (m_1, m_{10}, m_{16}) . The quartic $\langle X^\#, X^\# \rangle$ is the unique E_6 -invariant polynomial vanishing on the rank- ≤ 1 cone.*

Proof. On each finite graph, $\mathcal{C}$ is maximized at ω_n^* by Theorem 3.7. Under refinement, squared-distance penalties converge to $\text{Tr}(X^2)$ by the law of large numbers on μ ; the $\min_i$ selector enforces the quadratic radius constraint via a Lagrange multiplier, giving $[\text{Tr}(X^2) - c_2]^2$. The sharp term is forced by E_6 -invariance and the Freudenthal identity $X^\# \circ X = N(X)E$ (Appendix A.2): any E_6 -invariant quartic vanishing on rank ≤ 1 is proportional to $\langle X^\#, X^\# \rangle$. Independence of the refinement path follows from coercivity of V on compact E_6 . $\square$

4.3 Vacuum selection

Lemma 4.4 (Sharp locus). *$\langle X^\#, X^\# \rangle \geq 0$ with equality iff $\text{rank } X \leq 1$.*

Proof. In a Jordan frame $X = \text{diag}(\lambda_1, \lambda_2, \lambda_3)$, $X^\# = \text{diag}(\lambda_2\lambda_3, \lambda_3\lambda_1, \lambda_1\lambda_2)$. Positivity of diagonal entries gives $\langle X^\#, X^\# \rangle \geq 0$; zero occurs iff at most one $\lambda_i \neq 0$. $\square$

Theorem 4.5 (Global minimum on EIII). *For $\alpha, \beta, \gamma > 0$ and even $k \geq 2$, V is bounded below and coercive on compact E_6 . Every global minimizer lies on the rank-one primitive-idempotent orbit $\mathcal{M} = \text{EIII}$. The origin is a strict local maximum: $\text{Hess } V(0)$ is negative-definite.*

Proof. By Lemma 4.4, $\alpha\langle X^\#, X^\# \rangle$ is minimized on $\{\text{rank } X \leq 1\}$. The selector $\beta[\text{Tr}(X^2) - c_2]^2$ fixes the scale on each N -level set. Coercivity of V on the unitary 27 forces a minimum on a closed orbit; rank-one is minimal in the N -hierarchy. At $X = 0$, $\partial^2(\alpha\langle X^\#, X^\# \rangle)/\partial X^2$ is negative-definite in the transverse directions, yielding a saddle with negative Hessian. $\square$

Corollary 4.6 (Tree-level Goldstones). *At any vacuum point $\Phi_* \in \mathcal{M}$, $\text{Hess } V(\Phi_*)$ vanishes on $\mathfrak{m}$: all 32 coset modes are exact Goldstones at tree level.*

4.4 σ -model action

Let K be the H -invariant positive-definite metric on $\mathfrak{m}$ (Theorem 5.2). The tree-level action is

$$S[\Phi] = \int d^4x \sqrt{|g|} \left[\frac{1}{2} K_{AB} \nabla_\mu \Phi^A \nabla^\mu \Phi^B + V(\Phi) \right], \quad (4.2)$$

with $g_{\mu\nu}$ the soldered metric (Postulate 5.4). The quantum theory is defined by the Euclidean path integral $Z = \int \mathcal{D}\Phi e^{-S_E[\Phi]}$; OS reconstruction along the timelike clock field is Conjecture-level (Remark after Proposition 5.6).

4.5 Conditional universe fixed point

Definition 4.7 (Master data). Let $\mathfrak{U} = (\omega, \Phi, g_{\mu\nu}, \mathcal{H})$ denote capacity prices, order parameter, metric, and matter Hilbert space.

Theorem 4.8 (Conditional fixed-point structure). *Assume Postulates P1–P4. Then there exists data $\mathfrak{U}^*$ such that:*

- (i) ω^* maximizes $\mathcal{C}$ on $\mathcal{G}_{\text{Albert}}$;
- (ii) $\Phi_\star \in \mathcal{M}$ minimizes V ;
- (iii) $g_{\mu\nu}$ is the soldered Minkowski metric;
- (iv) $\mathcal{H}$ is the three-family SM Hilbert space of Theorem 7.5.

Uniqueness holds for (i)–(ii) by concavity and coercivity; (iii)–(iv) are unique up to gauge and family-basis choices.

Proof. (i)–(ii) are Theorems 3.7 and 4.5. (iii) is Postulate 5.4. (iv) follows from Theorem 7.5 under P2–P4 with anomaly cancellation Theorem 6.3. $\square$

5 Emergent Spacetime and Time

Four-dimensional Lorentzian geometry is not a consequence of the E_6 coset metric alone. Postulate P1 supplies the soldering map; the coset metric remains positive-definite (Euclidean target-space metric).

5.1 Coset metric and irreducibility

Theorem 5.1 (Real-irreducibility of $\mathfrak{m}$). *As a real H -module, $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ is irreducible of complex type: $\text{End}_H(\mathfrak{m}) \cong \mathbb{C}$.*

Proof. Complexifying, $\mathfrak{m}_{\mathbb{C}} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ with non-isomorphic summands exchanged by conjugation. A real submodule would be a complex submodule invariant under conjugation; Schur's lemma on each summand forbids nontrivial proper submodules. $\square$

Theorem 5.2 (Unique coset metric). *The restriction $K|_{\mathfrak{m}}$ of the Killing form is the unique H -invariant positive-definite inner product on $\mathfrak{m}$.*

Proof. H -invariant bilinear forms correspond to H -invariant maps $\mathfrak{m} \otimes \mathfrak{m} \rightarrow \mathbb{R}$. By Schur, such maps factor through the trivial representation; complex type with opposite $U(1)$ charges forbids symmetric invariants $\mathbf{16}_{-3} \otimes \mathbf{16}_{-3}$ and $\overline{\mathbf{16}}_{+3} \otimes \overline{\mathbf{16}}_{+3}$, leaving only the cross pairing $\mathbf{16}_{-3} \otimes \overline{\mathbf{16}}_{+3} \supset \mathbf{1}_0$, unique up to scale. $\square$

Remark 5.3. $K|_{\mathfrak{m}}$ is **positive-definite**, not Lorentzian. Signature $(-, +, +, +)$ is carried by the soldering form, not by the Killing restriction.

5.2 Postulate P1: soldering

The 2×2 Hermitian-octonionic block embeds as $H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$ with $\det$ the Minkowski norm; $\mathrm{SL}(2, \mathbb{O}) \cong \mathrm{Spin}(1, 9)$.

Postulate 5.4 (Soldering and dimension, P1). There exists a soldering map σ identifying four clock-field directions e_μ^a ($a = 0, 1, 2, 3$) with a four-dimensional subspace of $H_2(\mathbb{O})$ carrying $\eta_{ab} = \mathrm{diag}(-1, +1, +1, +1)$. The world manifold has $d = 4$. The vierbein emerges from the Maurer–Cartan $\mathfrak{m}$ -component (Section 6).

Proposition 5.5 (Emergent vierbein). *For a coset section $g(x) \in E_6$ with $g^{-1}\partial_\mu g = A_\mu + e_\mu$, the $\mathfrak{m}$ -component defines a composite vierbein. Under P1, $\eta_{\mu\nu} = e_\mu^a e_\nu^b \eta_{ab}$ has signature $(-, +, +, +)$.*

5.3 Emergent time

Proposition 5.6 (Page–Wootters clock). *In the semiclassical regime of induced gravity (Theorem 6.4), the Hamiltonian constraint $H_{\mathrm{phys}}|\psi\rangle = 0$ holds on physical states. The soldered timelike clock Φ^0 defines internal time; conditional evolution $U(\tau) = e^{-iH_\tau\tau}$ on physical states recovers standard quantum dynamics.*

Remark 5.7 (OS reconstruction). Euclidean correlators follow from (4.2). Lorentzian unitary dynamics requires Osterwalder–Schrader reconstruction along Φ^0 . **Conjecture:** the σ -model satisfies reflection positivity.

6 Gauge Sector and Induced Gravitation

6.1 Composite connection

The Lie algebra splits H -equivariantly as $\mathfrak{e}_6 = \mathfrak{h} \oplus \mathfrak{m}$ with $\mathfrak{h} = \mathfrak{so}_{10} \oplus \mathfrak{u}_1$ and $\mathfrak{m}$ as in (2.2).

Theorem 6.1 (Composite gauge connection). *For $g(x) \in E_6$, write $g^{-1}\partial_\mu g = \mathcal{A}_\mu + e_\mu$ with $\mathcal{A}_\mu \in \mathfrak{h}$, $e_\mu \in \mathfrak{m}$. Under $g \mapsto gh^{-1}$, $h \in H$,*

$$\mathcal{A}_\mu \mapsto h\mathcal{A}_\mu h^{-1} - (\partial_\mu h)h^{-1},$$

and the curvature $\mathcal{F}_{\mu\nu} = \partial_{[\mu}\mathcal{A}_{\nu]} + [\mathcal{A}_\mu, \mathcal{A}_\nu]$ transforms as an H field strength. All 46 generators of $\mathfrak{h}$ remain massless because H is unbroken.

Proof. Standard transformation of the canonical connection on the reductive homogeneous bundle $E_6 \rightarrow \mathcal{M}$. $\square$

Proposition 6.2 (Induced Yang–Mills). *At one loop, integrating out massive coset modes generates $S_{\mathrm{YM}} = \frac{1}{4\hat{g}^2} \int \mathrm{Tr}(\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu})$ with $\hat{g}^{-2} \propto f^2\mu^2/(16\pi^2)$, $[f] = \mathrm{mass}$.*

6.2 Anomaly cancellation

Anomalies are computed with Dynkin indices $A_{\mathbf{16}} = A_{\mathbf{10}} = 1$, $A_{\mathbf{1}} = 0$ and $U(1)$ charges from (2.3).

Theorem 6.3 (Green–Schwarz cancellation). *The cubic anomaly $\mathcal{A}_{U(1)^3}$ vanishes: $16 \cdot 1^3 + 10 \cdot (-2)^3 + 1 \cdot 4^3 = 0$. The mixed anomaly $\mathcal{A}_{U(1) \cdot \text{Spin}(10)^2}$ has coefficient -1 : $1 \cdot 1 + (-2) \cdot 1 + 4 \cdot 0 = -1$, and is cancelled by a Green–Schwarz term $\propto \text{tr}(F \wedge F) B$ with axion B from the $\mathbf{1}_{+4}$ sector.*

Proof. Direct arithmetic on charges and indices; see [6] for the mechanism. $\square$

6.3 Induced gravity

Theorem 6.4 (Sakharov induced gravity). *In a heat-kernel scheme, only massive modes contribute to the a_2 coefficient. With 28 physical massive coset scalars ($s = +1$, $m^2 = \mu^2$ canonically),*

$$M_{\text{Pl}}^2 = \frac{1}{16\pi^2} \cdot \frac{1}{6} \sum_{\text{massive}} s_i m_i^2 \log \frac{\Lambda^2}{m_i^2} \approx \frac{28}{6} \cdot \frac{\mu^2}{16\pi^2} \log \frac{\Lambda^2}{\mu^2} > 0. \quad (6.1)$$

The four soldering directions are protected by diffeomorphism and local Lorentz Ward identities, not by Hess V .

Remark 6.5 (Cosmological constant). The a_0 term yields an uncanceled Λ^4 vacuum energy. SJET does not solve the cosmological constant problem; Λ_{cc} is a tuned relevant coupling.

6.4 Renormalization group

Theorem 6.6 (Gauge β -function). *With canonical dimensionless coupling $\hat{g}^2$,*

$$\beta_{\hat{g}^2} = 2\hat{g}^2 + \frac{b_0}{48\pi^2} \hat{g}^4, \quad b_0 = \frac{11}{3} C_A - \frac{4}{3} T n_F - \frac{1}{6} T n_S. \quad (6.2)$$

For three chiral $\mathbf{16}s$ ($n_F = 3$) and $n_S = 28$ coset scalars:

$$b_0(3 \text{ gen}) = \frac{11}{3}(8) - \frac{4}{3}(2)(3) - \frac{1}{6}(2)(28) = \frac{88 - 24 - 28}{3} = 12. \quad (6.3)$$

The one-generation anchor ($n_F = 1$, $n_S = 2$) gives $b_0(1 \text{ gen}) = 26$. The gauge sector is asymptotically free ($\hat{g}_^2 = 0$).*

Conjecture 6.7 (Gravitational UV fixed point). *A truncated RG gives $\beta_\kappa = 2\kappa - (5/48\pi^2)\kappa^2$ with UV-attractive $\kappa_* = 96\pi^2/5$. This one-loop result is indicative, not rigorous [9].*

7 Standard Model Descent

Matter and electroweak symmetry breaking are **postulated** (P2–P4). Anomaly cancellation is a **theorem** once the **27** content is fixed.

7.1 Postulates P2–P4

Postulate 7.1 (Chiral matter, P2). Each family furnishes one chiral $\mathbf{16}$ of $\text{Spin}(10)$ with charges from (2.3). No mirror $\overline{\mathbf{16}}$ partners.

Postulate 7.2 (Three families, P3). Horizontal $SU(3)_F$ acts on $J_3(\mathbb{O}) \otimes \mathbb{C}^3$; the three factors are three chiral $\mathbf{16}s$. Three generations are *not* derived from Axioms I–III.

Postulate 7.3 (GUT–Higgs and Yukawa, P4). Breaking proceeds via $\mathbf{45}_H$ or $\mathbf{54}_H$ ($\text{Spin}(10) \rightarrow SU(5) \times U(1)_X$), $\mathbf{126}_H$ (breaks $B - L$), and $\mathbf{10}_H$ (electroweak). Yukawas $y_{ij} \mathbf{16}_i \mathbf{16}_j H$ generate fermion masses; type-I seesaw from $\mathbf{126}_H$ gives $m_\nu \sim m_D^2/M_R$.

Postulate 7.4 (Flavon hierarchy, P3'). Sequential $SU(3)_F$ breaking by flavon VEVs generates Yukawa hierarchy and CKM/PMNS mixing.

7.2 Descent chain

Theorem 7.5 (SM gauge structure under P4). *Assume P4. The breaking chain*

$$E_6 \xrightarrow{V} \text{Spin}(10) \times U(1) \xrightarrow{\mathbf{45}_H} SU(5) \times U(1)_X \xrightarrow{\mathbf{126}_H} SU(3)_c \times SU(2)_L \times U(1)_Y \xrightarrow{\mathbf{10}_H} SU(3)_c \times U(1)_{\text{em}}$$

yields the SM gauge group with GUT normalization $\alpha_1 = (5/3)\alpha_Y$.

Per family $\mathbf{16} = (q, u^c, d^c, \ell, e^c, \nu^c)$; $\mathbf{10}_H$ supplies $v = 246 \text{ GeV}$.

Remark 7.6 (Yukawa source term). A bare coupling $\bar{\psi} \Phi^a T_a \psi$ with Φ in $\{\mathbf{16}, \overline{\mathbf{16}}\}$ is *not* H -invariant and does not generate masses at the $\Phi_\star$ vacuum. Masses require P4 Higgs multiplets [7].

7.3 Scalar spectrum

Theorem 7.7 (Tree-level Goldstones). *At $\Phi_\star \in \mathcal{M}$, all 32 coset directions are massless at tree level (Corollary 4.6).*

Theorem 7.8 (Radiative degeneracy). *The Coleman–Weinberg potential [10] is H -invariant. Since $\mathfrak{m}$ is a single real-irreducible H -module (Theorem 5.1), Schur gives one eigenvalue on all 28 physical massive scalars; canonical mass $m^2 = \mu^2$.*

Remark 7.9 (Heterotic corroboration). The heterotic $\mathbb{Z}_5$ quintic gives $h^1(\hat{Q}, V_{16}) = 3$ [11]. **Conjecture**, not a derivation of P3.

8 Verification Ledger

This section records epistemic status and independent checks. The ledger is the referee-facing summary of what is proved, assumed, or conjectured.

8.1 Status table

8.2 Algebraic checks

Anomalies. Cubic: $16 - 80 + 64 = 0$. Mixed: $1 - 2 + 0 = -1$ (GS).

β -function. $b_0 = (88 - 24 - 28)/3 = 12$; one generation: $(88 - 8 - 4)/3 = 26$.

Dimensions. $45 + 1 + 16 + 16 = 78$; $1 + 10 + 16 = 27$; $16 + 16 = 32$; $28 + 4 = 32$.

Claim	Status	Check	Ref.
Stone representation	Theorem	Standard	§2.1
Hurwitz ladder	Theorem	dims 1, 2, 4, 8	3.1
1 10 16 partition	Prop.	Voronoi masses	2.7
Lift $\cong J_3(\mathbb{O})$	Theorem	27 coords	2.12
Orbit classification	Theorem	rank, N	2.9
Potential from $\mathcal{C}$	Theorem	limit	4.3
EIII vacuum	Theorem	coercivity	4.5
$\mathfrak{m}$ irreducible	Theorem	Schur	5.1
Unique coset metric	Theorem	charge pairing	5.2
Composite gauge	Theorem	Maurer–Cartan	6.1
$U(1)^3$ anomaly = 0	Theorem	$16 - 80 + 64$	6.3
Mixed GS anomaly	Theorem	coeff. -1	6.3
$M_{\text{Pl}}^2 > 0$	Theorem	28 scalars	(6.1)
$b_0(3) = 12$	Theorem	(6.3)	6.6
Asymptotic freedom	Theorem	$b_0 > 0$	6.6
CW mass degeneracy	Theorem	Schur	7.8
Soldering (1, 3), $d = 4$	Postulate	P1	5.4
Chiral 16 , 3 families	Postulate	P2–P3	§7.1
GUT–Higgs/Yukawa	Postulate	P4	7.3
Λ_{cc}	Postulate	tuned	§6.3
SM gauge (full masses)	Under P4	descent	7.5
$\kappa_* = 96\pi^2/5$	Conjecture	1-loop RG	6.7
OS positivity	Conjecture	σ -model	§5.3
Heterotic $h^1 = 3$	Conjecture	string	7.9

Table 1: Epistemic ledger for SJET.

8.3 Falsifiable predictions

F1. Proton decay. $\tau(p \rightarrow e^+\pi^0) \sim M_{\text{GUT}}^4/(\alpha_{\text{GUT}}^2 m_p^5)$; Super-K/Hyper-K constrain $M_{\text{GUT}} \gtrsim 10^{34}$ GeV.

F2. Neutrino mass scale. Seesaw with $M_R \sim 10^{14}$ GeV gives $m_\nu \sim 0.05$ eV.

F3. No fourth generation. A fourth **16** shifts b_0 and threatens asymptotic freedom.

F4. Coset mass degeneracy. One-loop CW mass μ on all 28 scalars; splittings only below M_{GUT} from H -breaking.

F5. R^2 bounds. If Conjecture 6.7 holds, higher-curvature couplings are suppressed in the UV.

9 Conclusions

Stone–Jordan Exceptional Theory is a *conditional* unification program: three axioms fix the discrete–algebraic arena and capacity dynamics; postulates P1–P4 complete the embedding into four-dimensional quantum field theory.

9.1 Proved from axioms (modulo standard mathematics)

- Identification of the capacity limit with the sharp-map potential (4.1).
- Vacuum selection on $\mathcal{M} = \text{EIII}$; origin is unstable.
- Irreducibility and unique metric on $\mathfrak{m}$.
- Composite $\text{Spin}(10) \times U(1)$ connection and anomaly cancellation.
- Induced $M_{\text{Pl}}^2 > 0$ from 28 massive scalars; $b_0(3) = 12$.

9.2 Postulated for SM contact

P1 (soldering), P2–P4 (matter, families, Higgs/Yukawa), and tuned Λ_{cc} .

9.3 Open problems

- O1.** Rigorous gravitational UV fixed point beyond one-loop truncation.
- O2.** OS reflection positivity for (4.2).
- O3.** Derive three generations without P3.
- O4.** Heterotic embedding of the exceptional cascade.
- O5.** Cosmological constant mechanism.

SJET is mathematically closed within its stated epistemic labels, internally consistent on anomaly and RG arithmetic, and verifiable via Table 1 and predictions F1–F5.

A Albert Algebra Identities

A.1 Matrix model

Elements of $J_3(\mathbb{O})$ are

$$X = \begin{pmatrix} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{pmatrix}, \quad \xi_i \in \mathbb{R}, \quad x_i \in \mathbb{O}, \quad (\text{A.1})$$

with Jordan product $X \circ Y = \frac{1}{2}(XY + YX)$. The trace inner product is $\langle X, Y \rangle = \text{Tr}(X \circ Y)$ and $Q(X) = \text{Tr}(X^2)$.

A.2 Sharp map and rank

The cubic norm is $N(X) = \det X$ (octonionic determinant). The sharp map is

$$X^\# = \text{adj}(X), \quad X^\# \circ X = N(X) E, \quad (X^\#)^\# = N(X) X, \quad (\text{A.2})$$

with $E = \text{diag}(1, 1, 1)$. In diagonal form $X = \text{diag}(\lambda_1, \lambda_2, \lambda_3)$, $X^\# = \text{diag}(\lambda_2\lambda_3, \lambda_3\lambda_1, \lambda_1\lambda_2)$. Thus $\text{rank } X \leq 1 \iff X^\# = 0$.

A.3 Freudenthal identity

For the quadratic adjoint $X \mapsto X^\#$,

$$(X \circ Y)^\# = (N(X)Y + N(Y)X - X^\# \circ Y^\#), \quad (\text{A.3})$$

which characterizes $J_3(\mathbb{O})$ among cubic vector spaces. Any E_6 -invariant quartic vanishing on $\{X^\# = 0\}$ is proportional to $\langle X^\#, X^\# \rangle$.

Acknowledgments

The author thanks prior UCFT/BAMLP development for the validated spine (archived in `old/`), especially the corrected β -function counts and anomaly arithmetic.

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